GM WORLDWIDE ENGINEERING STANDARDS	**Test Procedure Chassis**	**GMW16337**

Tire Dynamic Force and Moment

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This document describes the test equipment requirements, test methodology, and data format to supply tire force and moment (F&M) properties required to support nonlinear vehicle performance simulations for the development and validation of vehicles and vehicle subsystems.

1.2 Foreword.

1.2.1 The tire F&M properties are important to various vehicle simulations groups as tire performance properties contribute greatly to the vehicle's response to specific control inputs.

1.2.2 Tire F&M properties are a key element within the tire component technical specification (CTS) and in the validation of a specific tire design to those requirements.

1.2.3 The quantification of tire F&M properties requires the measurement of the net resultant forces and moments from stresses within the tire footprint area. This implies the footprint dynamics on the measurement equipment must be similar to those experienced under actual vehicle operation. Therefore, a flat surface machine using a textured surface is required for the quantification of tire F&M properties.

1.2.4 Since the tire F&M properties are dependent upon many factors including the measurement equipment, the test surface, environmental effects, and the tire history, the test procedure implementation and equipment suitability must be demonstrated prior to generating data for General Motors using this procedure. The suitability demonstration will not be a formal correlation, although that may become a future requirement, but will be established through an engineering assessment of the measurement results of a tire to this procedure with known performance attributes.

1.2.5 This procedure involves specific sequences of mechanical solicitations of the test tire with data collection under dynamic conditions. Specific sequences of slip angle, camber angle, and spindle torque solicitation of the test tire are available and may be selected to generate information to meet specific data requirements. Groupings of those specific sequences are referred to as test groups in Appendix A. The specific test groups to be run for each tire shall be established by the vehicle program team, considering the simulation requirements for the vehicle program and the test machine capabilities. See Appendix A for a list of specific test groups.

1.2.6 There is not a standard method of data analysis established for the data generated by this procedure. Therefore, the product of this procedure is a set of formatted data which is provided directly to General Motors. Included in this formatted data set will be the various transducer outputs and other information required to fully understand the data.

1.2.7 An industry standard reference tire, the ASTM E1136 SRTT, is used to quantify the frictional properties of the test surface. The SRTT is tested both before and after the acquisition of test tire data, to ensure the test surface properties are not unduly influencing the test results.

1.3 Applicability.

1.3.1 For all passenger cars, light truck tires, and some racing vehicle tires.

1.3.2 The tire properties generated from this procedure are to be determined using only new, unused, tires of the appropriate design level and conforming to all other tire CTS validation requirements unless fully discussed and agreed upon with the appropriate engineering personnel within General Motors.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

ASTM E1136	SAE J670e
ISO-8859-1	SAE J1987

2.2 GM Standards/Specifications.

None

3 Resources

3.1 Facilities. Not applicable.

3.2 Equipment. The equipment must be properly maintained and calibrated to traceable standards.

3.2.1 The equipment must provide a means of loading the tire against a flat, textured surface.

3.2.2 The equipment must be capable of steering the tire a specified amount about the Z-axis (slip angle) and the X-axis (camber angle).

3.2.3 Mechanical inputs during the test include the slip angle, camber angle, spindle torque, and radial load which are to be controlled through a closed-loop feedback system. The control system must demonstrate dynamic performance sufficient to achieve the required input magnitudes and rates.

3.2.4 The control system must continuously monitor the test in a manner to prohibit damage to the test equipment in the event of a sudden loss of inflation pressure. The equipment must be capable of reliably allowing for rim to test surface clearances of 20 mm before aborting the test condition. This limit tripping must be a soft limit that will remove the load from the test tire without aborting the entire test as this is an expected occurrence within this procedure.

3.2.5 The roadway control system must be capable of dynamically controlling the speed, direction, and tracking of the roadway.

3.2.6 The roadway must be rigidly and evenly supported to react the vertical forces generated within the tire footprint. The support structure must be oriented parallel to the spin axis of the test spindle and perpendicular to the wheel plane at zero degrees of inclination angle.

3.2.7 The test surface typically consists of an abrasive material glued to a steel belt. The typical abrasive material is 120-grit 3M-ite. This product is used to simulate the texture of a typical road surface. In order to more closely represent the road surface, the surface is conditioned through stoning process and broken-in prior to the initial use. See Appendix G for the details regarding new surface preparation.

3.2.8 The transducer system must measure the six forces and moments generated by the tire as well as the slip and camber angles, the loaded radius, spindle and roadway speeds, and various temperatures. See Table 1 below in Section 5.

3.2.9 The data acquisition system must acquire the output from the transducers and ultimately produce outputs presented as scaled engineering units.

3.2.10 The data acquisition method utilized for this procedure will be time-based at a rate of at least 250 Hz, with simultaneous sampling across all channels.

3.2.11 The full scale equipment capability values are provided in Appendix I. These values are to be considered general guidelines in determining the suitability of a given piece of equipment to execute this procedure. Actual equipment demands during a test are often a combination of the various limits which cannot be adequately comprehended within the scope of this procedure.

The equipment meeting the minimum performance levels described within Table H1 will need to be used in the execution of this procedure for many tires. The equipment described in Table H2 may not exist as of the writing of this procedure. However, equipment meeting those performance levels will be needed to execute this procedure for the larger passenger car and light truck tires used within General Motors.

The data required to support General Motors product development is being gathered today through the combination of MTS Flat-Trac and Calspan equipment. None of this equipment is fully capable per Table H2, but has provided a data from which tire performance estimates can be extrapolated from to support vehicle development.

3.3 Test Vehicle/Test Piece.

3.3.1 Test Specimen. Test tires must be free of all tags, tape, stones, or any other foreign material in the tread area and be in new, unused condition.

3.3.2 Test Wheel. The wheel must be selected properly to facilitate correct test results. The following should be considered when selecting the proper test wheel.

3.3.2.1 The wheel width must match that requested on the test request.

3.3.2.2 The wheel must have the proper rim flange contour for the test tire.

3.3.3 Test Surface. The roadway surface must be free of foreign material including, but not limited to, oils, rubber dust, or other deposits from previous usage.

3.3.4 Sample Size. Consideration must be given to the number of tires to be tested, or sample size, for a specific purpose. The complete data set from this procedure will require a minimum of nine tires to be tested under varying conditions. It is recommended additional tires be available to facilitate additional tests in the event of unforeseen testing events. This procedure applies abusive test conditions to the test specimens, rendering them unsuitable for additional use or repeated testing.

3.4 Test Time.

Calendar time:	2.0 days (8 h shifts)
Test hours:	13.5 hours
Coordination hours:	2.0 hours

3.5 Test Required Information.

3.5.1 The following information regarding the tire must be provided in order to facilitate the test.

- Construction number
- Tire size
- Service description
- Supplier (manufacturer)
- Load rating

3.5.2 The following information related to the desired test conditions must also be specified.

3.5.2.1 Rim width shall be the expected production wheel width.

3.5.2.2 Load will often be referenced as a percentage of the "100% Test Load". The 100% Test Load will be related to the normal load of the intended production vehicle. For example, the average front wheel load of the intended production vehicle, in its normal load ballast condition, may be used as the "100% Test Load". This load value will need to be obtained from the appropriate General Motors engineer.

3.5.2.3 Inflation pressure will be the warm operating inflation pressure for the intended application vehicle. This will typically be 20 kPa above the placard pressure for passenger car tires and vehicles and 35 kPa above placard for light truck tires and vehicles.

3.6 Personnel/Skills.

3.6.1 Test operators must be trained in the safe operation of the measurement equipment. The measurement equipment, by necessity, involves moving parts and loading mechanisms which must be understood and operated using proper safety precautions.

3.6.2 Operators must be trained in the selection of test components and fixtures to facilitate the proper testing of the tires.

3.6.3 Operators must be trained to observe, and correct when necessary, the test surface condition. The surface condition and contamination level directly impact the quality of the test results.

4 Procedure

4.1 Preparation.

4.1.1 Thermally stabilize the test tires by storing in an area with an ambient temperature within 5 °C of the test machine environment for at least 24 h prior to testing.

4.1.2 Mount the test tire on the test wheel.

4.1.2.1 Follow approved mounting procedures, such as those published by the Rubber Manufacturer's Association, when mounting the test tire onto test wheels.

4.1.2.2 Inspect the tire bead and the bead seat of the wheel to insure this interface is free of damage, defects, and foreign material. Clean or replace test pieces as necessary.

4.1.2.3 Use only a thin film of an approved mounting lubricant to facilitate proper tire mounting and bead seating. Excessive mounting lubricant can cause slippage at the tire and rim interface during testing.

4.1.2.4 Inspect the bead area of the mounted tire and wheel assembly to ensure the tire bead is firmly seated on the wheel.

4.2 Conditions.

4.2.1 Environmental Conditions. The ambient temperature in the room where the measurement equipment is operating shall be controlled to 21 ± 3 °C to minimize the thermal gradients within the test equipment and instrumentation. The room temperature gradients should also be within the equipment manufacturer recommendations which will typically be less than 1 °C/h.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 The specific test conditions for each tire to be tested to obtain the complete data set are provided in Appendix A or Appendix B for the SRTT reference tire.

4.3 Instructions.

4.3.1 Quantify the Surface Condition. Quantify the frictional properties of the test surface using an ASTM E1136 SRTT and the conditions provided in Appendix B.

4.3.1.1 Perform instructions in paragraphs 4.3.2 thru 4.3.3.

4.3.1.2 If the tire is a new tire which has not previously been prepared for use according to Appendix B, then conduct the break-in process described in Appendix B. Note the break-in procedure for the ASTM E1136 SRTT used for surface friction properties is different than other test tires.

4.3.1.3 Acquire the surface frictional properties by performing the instructions in paragraph 4.3.2.4, including the subparagraphs, to gather a total of ten braking slip ratio sweeps.

4.3.2 Perform Tire Test. Tire testing will be completed in groups according to those found in Appendix A. The specific conditions for that test group will be listed in that section.

4.3.2.1 Install Test Assembly. Install the tire and wheel assembly onto the test equipment.

4.3.2.2 Inflate Tire. Regulate the inflation pressure.

4.3.2.3 Perform Break-In.

4.3.2.3.1 Apply a nominal load to the test assembly.

4.3.2.3.2 Set the slip angle to 0 degrees.

4.3.2.3.3 Set the inclination angle to degrees.

4.3.2.3.4 Set the test speed.

4.3.2.3.5 Set the test load to the 100% Test Load.

4.3.2.3.6 Regulate the inflation pressure to the test inflation pressure.

4.3.2.3.7 Continue the operation under the above conditions for 60 s.

4.3.2.3.8 Cap the inflation pressure.

4.3.2.3.9 Ramp the slip angle from 0 degrees to –0.5 degrees at 4 degrees/s while maintaining the applied load.

4.3.2.3.10 Dwell at -0.5 degrees for 0.25 s.

4.3.2.3.11 Ramp the slip angle from -0.5 degrees to +20 degrees at 4 degrees/s while maintaining the applied load.

4.3.2.3.12 Dwell at +20 degrees for 0.25 s.

4.3.2.3.13 Ramp the slip angle from +20 to 0 degrees at 10 degrees/s.

4.3.2.3.14 Regulate the inflation pressure.

4.3.2.3.15 Continue the operation under the above conditions for 60 s.

4.3.2.3.16 Cap the inflation pressure.

4.3.2.3.17 Ramp the slip angle from 0 degrees to +0.5 degrees at 4 degrees/s while maintaining the applied load.

4.3.2.3.18 Dwell at +0.5 degrees for 0.25 s.

4.3.2.3.19 Ramp the slip angle from +0.5 to -20 degrees at 4 degrees/s while maintaining the applied load.

4.3.2.3.20 Dwell at -20 degrees for 0.25 s.

4.3.2.3.21 Ramp the slip angle from -20 to 0 degrees at 10 degrees/s.

4.3.2.3.22 Regulate the inflation pressure.

4.3.2.3.23 If the tire is intended to be used for testing including spindle torque application, then perform the addition break-in sequences detail in paragraphs 4.3.2.3.24 to 4.3.2.3.34. If this tire is not to include spindle torque application, then proceed to paragraph 4.3.2.4.

4.3.2.3.24 Set the slip angle to 0 degrees.

4.3.2.3.25 Set the inclination angle to 0 degrees.

4.3.2.3.26 Set the test speed.

4.3.2.3.27 Set the test load to the 100% Test Load.

4.3.2.3.28 Continue the operation under the above conditions for 20 s.

4.3.2.3.29 Cap the inflation pressure.

4.3.2.3.30 Ramp the slip ratio from 0 to -80% at a slip ratio rate of 50%/s.

4.3.2.3.31 Dwell at -80% slip ratio for 0.1 s.

4.3.2.3.32 Ramp the slip ratio from -80 to 0% at a slip ratio rate of 50%/s.

4.3.2.3.33 Regulate the inflation pressure.

4.3.2.3.34 Repeat 4.3.2.3.24 to 4.2.3.33 for a total of 25 iterations.

4.3.2.4 Perform Dynamic Sweep Sequence.

4.3.2.4.1 Operate the test assembly at the 100%. Load, zero degrees inclination angle, zero degrees slip angle, zero applied spindle torque, and at the stated test speed for 60 s.

4.3.2.4.2 Cap the inflation pressure.

4.3.2.4.3 Apply all of the steady-state test conditions to the test assembly as quickly as possible. This step should be completed within 1 to 2 s and is dependent on the response rate of the test equipment.

4.3.2.4.4 If this is a slip ratio test, test groups 9 to 13 in Appendix A, proceed to 4.3.2.4.7.

4.3.2.4.5 Initiate the dynamic control channel by sweeping the command signal, using the appropriate sweep rate, to -0.5 degrees.

4.3.2.4.6 Dwell at this dynamic control command level for 0.2 s.

4.3.2.4.7 Sweep the dynamic control channel, using the appropriate sweep rate, to the maximum value for this test sequence.

4.3.2.4.8 Dwell at the maximum value for the dynamic control channel for 0.2 s.

4.3.2.4.9 Remove the input from the dynamic control channel by sweeping the command signal, using the appropriate sweep rate to zero.

4.3.2.4.10 Dwell at the zero command for the dynamic control channel for 0.2 s.

4.3.2.4.11 Reset all of the steady-state control channels to their nominal set points.

4.3.2.4.12 Regulate the inflation pressure.

4.3.2.4.13 Operate the test assembly at the nominal test conditions for 60 s.

4.3.2.4.14 Cap the inflation pressure.

4.3.2.4.15 Apply all of the steady-state test conditions to the test assembly as quickly as possible.

4.3.2.4.16 If this is a slip ratio test, proceed to 4.3.2.4.19.

4.3.2.4.17 Initiate the dynamic control channel by sweeping the command signal, using the appropriate sweep rate, to +0.5 degrees.

4.3.2.4.18 Dwell at this dynamic control command level for 0.2 s.

4.3.2.4.19 Sweep the dynamic control command signal, using the appropriate sweep rate, to the minimum value for this test sequence.

4.3.2.4.20 Dwell at the minimum value for the dynamic control channel for 0.2 s.

4.3.2.4.21 Remove the input from the dynamic control channel by sweeping the command signal, using the appropriate sweep rate to zero.

4.3.2.4.22 Dwell at the zero command for the dynamic control channel for 0.2 s.

4.3.2.4.23 Reset all of the steady-state control channels to their nominal set points.

4.3.2.4.24 Regulate the inflation pressure.

4.3.2.4.25 Continue with paragraph 4.3.2.4.1 above using the next set of test conditions for the current test assembly until all test sequences for this test assembly have been completed.

4.3.2.4.26 Select the next test assembly and begin with paragraph 4.3.2.1 repeating until all test sequences have been completed.

4.3.3 Quantify the Surface Condition. Repeat paragraph 4.3.1 above at the conclusion of the testing for each test tire group.

4.3.4 Test Tire Disposal. The tires used in this testing are to be rendered unusable by cutting the bead bundle, splitting the tire circumferentially, or other similar means.

5 Data

5.1 Sampling and Filtering.

5.1.1 Data are to be collected and written to data files during the active sections of the testing. Data are to be reported from 4.3.2.4.4 to 4.3.2.4.10 and from 4.3.2.4.16 to 4.3.2.4.22. The data acquisition system may be inactive during the machine operation outside these steps.

5.1.2 The sampling rate shall be 250 Hz or greater using simultaneous sample and hold or bursting techniques. If the bursting technique is used, then the burst sample rate must be at least an order of magnitude higher than the base sample rate.

5.1.3 Anti-alias filtering techniques appropriate to the machine design and this procedure are expected to be employed.

5.1.4 All recorded data signals are expected to utilize the same filters to eliminate phasing of signals due to filter induced phase lags.

5.1.5 The following table provides a list of the required data channels.

Table 1: Required Channel List

Channel Name	Description	Unit
ET	Elapsed Time	seconds
SA	Lateral Slip Angle	degrees
IA	Inclination Angle	degrees
V	Belt Speed	km/h
N	Wheel Rotational Speed	rpm
P2	Inflation Pressure	kPa
FX	Tire Longitudinal Force	N
FY	Tire Lateral Force	N
FZ	Tire Vertical Force	N
MX	Tire Overturning Moment	Nm
MY	Tire Drive/Brake Torque	Nm
MZ	Tire Aligning Moment	Nm
SL	Longitudinal Slip Ratio (SAE)	m/m
RL	Actual Tire Radius	mm
TSTC	Tread Surface Temperature (Center 1/3)	degrees
TSTI	Tread Surface Temperature	degrees

	(Inner 1/3)	
TSTO	Tread Surface Temperature (Outer 1/3)	degrees
RST	Road Surface Temperature	degrees
AMBTMP	Ambient Temperature	degrees

5.2 Calculations.

5.2.1 The raw data analysis detailed below assumes the data are either acquired or has been preprocessed such that the following statements regarding the raw data, or machine generated test results, are true.

5.2.1.1 Data are represented in SI units with appropriate tare measurements subtracted and gain factors applied.

5.2.1.2 Data sign conventions are consistent with the SAE tire coordinate system as described in SAE J670e.

5.2.1.3 The data were collected using the data acquisition method described in the Sampling and Filtering section above.

5.2.1.4 Data were collected while the equipment control system had control of the test and the machine response was sufficient such that the error between the command signals and the test conditions realized was small.

5.2.2 Channel SL is calculated as shown below. The values of V_1 and N_1 are determined when MZ = 0.0 Nm and SA = 0.0 degrees, otherwise known as free-rolling conditions,

$$SL = \left[\left(\frac{V_1}{N_1} \right) * \left(\frac{N}{V} \right) \right] - 1$$

5.3 Interpretation of Results. Not applicable.

5.4 Test Documentation. All electronic data file formats are defined by the Global Tire-Wheel Test System (GTWTS). The file format details can be obtained from the global test coordinator for Tire-Wheel Systems or referenced from the GM Tire-Wheel Systems internal web site. All previous test data file formats are now obsolete.

When a test request is submitted in GTWTS, the tester will receive via email a copy of the Test Request in PDF form, input files and datasheets (as applicable), and XML Skeleton Data files. Electronic results files (TPC or XML formats) should be submitted for loading into the GTWTS and for generating the test report from the system.

5.4.1 A sample XML file is shown in Appendix C. The name of the file will be a concatenation of six pieces of information, or abbreviations, representing the source and content of the information. An explanation of the file naming convention is also available in Appendix C.

5.4.2 File attachments are used to load the majority of the information generated from this procedure into the GTWTS, the final repository. There are 3 types of files to be attached to the XML file.

5.4.2.1 The first file attachment is a test log as shown in Appendix D. The test log includes information allowing General Motors to quickly understand the contents of the data files and significant events occurring during the physical testing of the tires.

The file name will be similar to that defined for the xml file in 5.4.1, except the sequence number will be replaced by the label "TESTLOG" immediately prior to the file extension, and the file extension will be appropriate for the type of file, such as "xls" for a Microsoft Excel file.

5.4.2.2 The second file attachment is a tire information log detailing the individual tire data such as shown in Appendix E. The data for this information log should be read directly from the test specimens as it is intended to provide independent confirmation the correct tires were tested.

The file name will be defined similar to the xml file in 5.4.1, except the sequence number will be replaced by the label "TIREIDLOG" immediately prior to the file extension, and the file extension will be appropriate for the type of file, such as "xls" for a Microsoft Excel file.

5.4.2.3 The third file type contains the time history data measurements acquired during each prescribed test. These data files are to be formatted to appear as the sample shown in Appendix F.

The file name will be defined similar to the xml file in 5.4.1, except the sequence number will be replaced by the label "TESTXX" immediately prior to the file extension. The file extension for this file type will be "dat". The "XX" will be the data file number associated with the measurement sequence in Appendix A.

This is an ASCII text file and is not to contain any characters not represented by the ASCII Character

Codes within the range of 000 to 126. More detail will be provided in Appendix F.

5.4.2.3.1 The file shall contain only those characters producing alphanumeric characters, plus and minus symbols.

5.4.2.3.2 The line termination character sequence is to be the carriage return/line feed combination, otherwise known as the ASCII Character Codes 013 and 010, respectively.

5.4.2.3.3 The name:value pairs in the header section of the file are to be separated by the colon character, ASCII Character Code 058, and the labels are to be duplicated exactly as shown in the sample output, Figure F1 in Appendix F, except the first line is a free character field to contain the name of the test site.

5.4.2.3.4 The ET column is to contain the elapsed time since the current test tire's testing began.

5.4.2.3.5 The data values within the time history portion of the file are to be separated by the space character, or ASCII Character Code 32.

5.4.2.3.6 There will be one data file created for each group of testing identified in Appendices A and B.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.1 Safety Precautions.

6.1.1 This standard involves the use of test equipment with rotating components and multiple hydraulically controlled mechanisms. Operators are to be aware of those hazards and take reasonable precautions to prevent personal injury. Below are several notable areas where caution should be exercised. However, it is not intended to be an exhaustive list.

6.1.2 Make sure the roadway has come to a complete stop before approaching the machine to change the tire and wheel assembly, spindle adapters, etc.

6.1.3 Inspect the test wheels for bent rims or excessive galling in the nut boss area.

6.1.4 Inspect fasteners for excessive wear or deformation.

6.1.5 Torque fasteners to specification when installing the tire and wheel assembly. A torque between 110 to 125 Nm is recommended if not otherwise specified.

6.1.6 Do not lift large tire and wheel assemblies, such as light truck assemblies, onto the machine unassisted. A lifting device should be used for this purpose.

7 Notes

7.1 Glossary.

Slip Ratio: Relative slip rate between the test tire and the component of the belt velocity coincident with the heading of the test wheel.

7.2 Acronyms, Abbreviations, and Symbols.

CTS	Component Technical Specification
DOT	United States of America's Department of Transportation
F&M	Force and Moment
GTWTS	Global Tire-Wheel Test System
SA	Slip Angle
SR	Slip Ratio
SRTT	ASTM E1136, P195/75R14 Standard Reference Test Tire
TIP	Test Improvement Process
T&RA	Tire and Rim Association
TWS	Tire-Wheel Systems

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW16337

9 Release and Revision

This standard was originated in December 2009. It was first approved by the Tire Global Subsystem Leadership Team in July 2010. It was first published in July 2010.

Issue No.	Publication Date	Description (Organization)
1	JUL 2010	Initial publication.

Appendix A

Table A1: Test Conditions

Test Group 1
Tire No.1
Data File No. 01

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SA Rate (Degrees/Second)
100	0	± 20	0	4
200	0	± 20	0	4
150	0	± 20	0	4
60	0	± 20	0	4
30	0	± 20	0	4
250	0	± 20	0	4
300 Note 1	0	± 20	0	4

Test Group 2
Tire No. 1
Data File No. 02

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	IA Rate (Degrees/Second)
100	± 8	0	0	1
200	± 8	0	0	1
150	± 8	0	0	1
60	± 8	0	0	1
30	± 8	0	0	1
250	± 8	0	0	1
300 Note 1	± 8	0	0	1

Test Group 3
Tire No. 2
Data File No. 03

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SA Rate (Degrees/Second)
100	+4	± 20	0	4
150	-4	± 20	0	4
60	+4	± 20	0	4
200	-4	± 20	0	4
250	+2	± 20	0	4
300 Note 1	+2	± 20	0	1
100	0	± 15	0	4

Test Group 4
Tire No. 2
Data File No. 04

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	IA Rate (Degrees/Second)
100	-4	± 20	0	4
150	+4	± 20	0	4
60	-4	± 20	0	4
200	+4	± 20	0	4
250	-2	± 20	0	4
300 Note 1	-2	± 20	0	4

Test Group 5
Tire No. 3
Data File No. 5

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SA Rate (Degrees/Second)
100	-2	± 20	0	4
150	+2	± 20	0	4
60	+8	±20	0	4
200	-8	± 20	0	4
250	-8	± 20	0	4
300 Note 1	-8	± 20	0	4
100	0	± 15	0	4

Test Group 6
Tire No. 3
Data File No. 06

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	IA Rate (Degrees/Second)
100	2	± 20	0	4
150	-2	± 20	0	4
60	-8	± 20	0	4
200	8	± 20	0	4
250	-4	± 20	0	4
300 Note 1	-4	± 20	0	4

Test Group 7
Tire No. 4
Data File No. 07

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SA Rate (Degrees/Second)
60	-2	± 20	0	4
200	+2	± 20	0	4
100	-8	± 20	0	4
150	+8	± 20	0	4
250	+8	± 20	0	4
100	0	± 15	0	4
300 Note 1	+8	± 20	0	4

Test Group 8
Tire No. 4
Data File No. 08

Speed = 80 km/h (50 mph)
Test Type = Combined Slip and Camber

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	IA Rate (Degrees/Second)
60	+2	± 20	0	4
200	-2	± 20	0	4
100	+8	± 20	0	4
150	-8	± 20	0	4
250	+4	± 20	0	4
300 Note 1	+4	± 20	0	4

Test Group 9
Tire No. 5
Data File No. 09

Speed = 55 km/h (35 mph)
Test Type = Pure Braking

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
60	0	0	-80	50
80	0	0	-80	50
100	0	0	-80	50
130	0	0	-80	50
160	0	0	-80	50
200	0	0	-80	50
250	0	0	-80	50
300 Note 1	0	0	-80	50

Test Group 10
Tire No. 6
Data File No. 10

Speed = 55 km/h (35 mph)
Test Type = Combined Braking and Cornering

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
60	0	-4	-80%	50
60	0	+	-80%	50
100	0	-4	-80%	50
100	0	+	-80%	50
60	+2	0	-80%	50
60	-2	0	-80%	50
100	4	0	-80%	50
100	-4	0	-80%	50
100	0	-16	-80%	50
100	0	+6	-80%	50

Test Group 11
Tire No. 7
Data File No. 11

Speed = 55 km/h (35 mph)
Test Type = Combined Braking and Cornering

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
100	-2	0	-80%	50
100	+2	0	-80%	50
60	0	-2	-80%	50
60	0	+2	-80%	50
100	-8	0	-80%	50
100	+8	0	-80%	50
60	-4	0	-80%	50
60	+4	0	-80%	50
60	0	-16	-80%	50
60	0	+16	-80%	50

Test Group 12
Tire No. 8
Data File No. 12

Speed = 55 km/h (35 mph)
Test Type = Combined Braking and Cornering

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
60	0	-8	-80%	50
60	0	+8	-80%	50
100	0	-2	-80%	50
100	0	+2	-80%	50
60	+8	0	-80%	50
60	-8	0	-80%	50
100	0	-8	-80%	50
100	0	+8	± 80%	50

Test Group 13
Tire No. 9
Data File No. 13

Speed = 55 km/h (35 mph)
Test Type = Combined Braking and Cornering

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
100	-4	-4	-80%	50
100	+4	+4	-80%	50
100	-4	+4	-80%	50
100	+4	-4	-80%	50
60	-8	-8	-80%	50
60	+8	+8	-80%	50
60	-8	+8	-80%	50
60	+8	-8	-80%	50
60	-4	-4	-80%	50
60	+4	+4	-80%	50
60	-4	+4	-80%	50
60	+4	-4	-80%	50
100	-8	-8	-80%	50
100	+8	+8	-80%	50
100	-8	+8	-80%	50
100	+8	-8	-80%	50

Note 1: The 300% load is only run for light truck applications and where the machine capability exists to achieve the test inputs required to run this test condition.

Appendix B

Table B1: SRTT Usage Test Conditions and Break-In

First Data File No. C01
Second Data File #C02 (Data File Sequences will continue to include all required control tire runs.)
100% Load = 4580 N (1031 lb)
Inflation Pressure = 241 kPa (35 psi)
Speed = 55 km/h (35 mph)

Test Type = Pure Braking

Load (%)	IA (Degrees)	SA (Degrees)	SR (%)	SR Rate (%/Second)
60	0	0	-80	50
80	0	0	-80	50
100	0	0	-80	50
130	0	0	-80	50
160	0	0	-80	50
200	0	0	-80	50
250	0	0	-80	50

B1 Purpose. This process will condition a new ASTM E1136 SRTT, preparing it for use as a control tire.

B1.1 A new SRTT must be cycled a number of times to ensure measurement reproducibility as the tire will be tested repeatedly over time to determine and track the frictional properties of the test surface used to quantify other test tires. The new SRTT performance must be stabilized so the tire evolution effects are not attributed to test surface changes.

B2 All of the instructions and test conditions not specifically called out in the following paragraphs associated with this procedure are to be applied during the New SRTT Conditioning.

B3 Instructions.

B3.1 Set the slip angle to 0 degrees.

B3.2 Set the inclination angle to 0 degrees.

B3.3 Set the test speed.

B3.4 Set the test load to the 100% Test Load.

B3.5 Continue the operation under the above conditions for 20 s.

B3.6 Cap the inflation pressure.

B3.7 Ramp the slip ratio from 0% to -80 at a slip ratio rate of 50%/s.

B3.8 Dwell at -80% slip ratio for 0.2 s.

B3.9 Ramp the slip ratio from -80% to 0% at a slip ratio rate of 50%/s.

B3.10 Regulate the inflation pressure.

B3.11 Repeat B.3.1 to B.3.11 for a total of 10 iterations.

B3.12 Calculate the peak braking coefficient for each of the 10 braking torque applications.

B3.12.1 Determine the evaluation window width;

$$w = \text{int}\left[\frac{3 \times Rate_{DAC}}{100}\right]$$

Where:

w	= Evaluation window width
***Rate**$_{DAC}$*	= Data acquisition sampling rate [Hz]
int	= Mathematical integer function

B3.12.2 Calculate the braking coefficient for each point in the data set of interest a;

$$c_i = \sum_{i+(w/2)}^{i-(w/2)} \frac{Fx_i}{Fz_i}$$

Where: for each point,

c_i	= Braking coefficient
w	= Evaluation window width
Fx_i	= Longitudinal force, or braking force [N]
Fz_i	= Applied normal load [N]

B3.13 Calculate the mean and standard deviation of the 10 braking torque applications above.

B3.14 Calculate a second mean of those braking coefficients using only the values inside the range defined by the mean ± 1.0 standard deviation.

B3.15 Evaluate the stability of the calculated second mean values.

B3.16 If the values are stable, then the tire is ready for use as within this procedure. If the values are not stable, then repeat steps under paragraph B3.

The values are considered stable when they are trending by less than 0.01 per measurement group.

Appendix C: Global Tire Wheel Test System Example XML Data File

The file name is a combination of information provided by the Global Tire and Wheel Test System (GTWTS) coordinator. The separate segments of the file name are to be separated by underscores, ASCII Code 95. For the example below,

GM	GM is the “Tester” for the information to be provided in the xml file.
T09-05586	This character sequence represents the assigned “Test Request Number” from the GTWTS.
DFM	The three character code associated with “Dynamic Force and Moment” test area within GTWTS.
001	A three digit “File Sequence Number” assigned by the file originator to create unique file names to be provided to the GTWTS.
Filename	GM_T09-05586_DFM_001.xml

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<tpc:tpc_template xmlns:tpc="http://www.gtwts.gm.com/tpc">
    <tpc:validation_data>
        <tpc:request_number>T09-05586</tpc:request_number>
        <tpc:pqms_per_number>2009-06-01-53956</tpc:pqms_per_number>
        <tpc:test_obj_part_number>5D1456Q</tpc:test_obj_part_number>
        <tpc:supplier_lv_id>GDY</tpc:supplier_lv_id>
        <tpc:tire_size displayName="Tire Size">P265/70R17</tpc:tire_size>
        <tpc:load_index displayName="Load Index">113</tpc:load_index>
    </tpc:validation_data>
    <tpc:tester_code>GMNA-TWS</tpc:tester_code>
    <tpc:site_code>CAL-BNY</tpc:site_code>
    <tpc:technician><![CDATA[DAVID HOWLAND]]></tpc:technician>
    <tpc:end_test_date>17-DEC-2009</tpc:end_test_date>
    <tpc:machine_name><![CDATA[TIRF]]></tpc:machine_name>
    <tpc:machine_id><![CDATA[1]]></tpc:machine_id>
    <tpc:no_of_tires displayName="Number of Tires">9</tpc:no_of_tires>
    <tpc:test_number>5586</tpc:test_number>
    <tpc:processing_sw_name><![CDATA[N/A]]></tpc:processing_sw_name>
    <tpc:processing_sw_version><![CDATA[N/A]]></tpc:processing_sw_version>
    <tpc:comments displayName="Comments"><![CDATA[NONE]]></tpc:comments>
    <tpc:no_of_attachments>
        <tpc:no_of_iterations>17</tpc:no_of_iterations>
        <tpc:file_name>GM_T09-05586_TESTLOG.XLS</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TIREIDLOG.XLS</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST00.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST01.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST02.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST03.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST04.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST05.DAT</tpc:file_name>
        <tpc:file_name>GM_T09-05586_TEST06.DAT</tpc:file_name>
```

<tpc:file_name>GM_T09-05586_TEST07.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST08.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST09.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST10.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST11.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST12.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST13.DAT</tpc:file_name>
<tpc:file_name>GM_T09-05586_TEST14.DAT</tpc:file_name>
</tpc:no_of_attachments>

</tpc:tpc_template>

Appendix D

Figure D1: Sample Log

Run Number	Data Filename	Tire ID	Surface Decription	Inflation [kPa]*	Speed [km/h]	Slip Ratio [%]	Slip Angle [Degrees]	Inclination [Degrees]	Load [N]	Testing Comments
T09-5586-0	GM_T09-05586_DFM_Data_000.dat	SRTT-AN2209-127	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-12	L1	W1,C1
T09-5586-1	GM_T09-05586_DFM_Data_001.dat	5586-01	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-10	L1	W1
T09-5586-2	GM_T09-05586_DFM_Data_002.dat	5586-01	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-8	L1	W1
T09-5586-3	GM_T09-05586_DFM_Data_003.dat	5586-02	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-6	L1	W1
T09-5586-4	GM_T09-05586_DFM_Data_004.dat	5586-02	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-4	L1	W1
T09-5586-5	GM_T09-05586_DFM_Data_005.dat	5586-03	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-2	L1	W1
T09-5586-6	GM_T09-05586_DFM_Data_006.dat	5586-03	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	0	L1	W1,C4
T09-5586-7	GM_T09-05586_DFM_Data_007.dat	5586-04	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	2	L1	W1
T09-5586-8	GM_T09-05586_DFM_Data_008.dat	5586-04	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	4	L1	W1
T09-5586-9	GM_T09-05586_DFM_Data_009.dat	5586-05	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	6	L1	W1
T09-5586-10	GM_T09-05586_DFM_Data_010.dat	5586-06	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	8	L1	W1,C2
T09-5586-11	GM_T09-05586_DFM_Data_011.dat	5586-07	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	10	L1	W1
T09-5586-12	GM_T09-05586_DFM_Data_012.dat	5586-08	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	12	L1	W1
T09-5586-13	GM_T09-05586_DFM_Data_013.dat	5586-09	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	15	L1	W1
T09-5586-14	GM_T09-05586_DFM_Data_014.dat	SRTT-AN2209-127	120, 3-Mite (GM Spec)	38 c	80	Free-Rolling	S1	-12	L1	W1,C3

(*) Inflation Pressure Key: c = Capped / r = Regulated / v = Vented

Speed Comment Key:

Slip Ratio Comment Key:

Slip Angle Comment Key:

S1: +0.5 Degrees -> -25.0 Degrees, -0.5 Degrees -> +25.0 Degrees

Inclination Comment Key:

Load Comment Key:

L1: Normal Load Sequence, 371 N, 1113 N, 1855 N, 2597 N, 3338 N, 4080 N, 4822 N, 5564 N

L2: Normal Load Sequence, 5564 N

Testing Comments Key:

W1: Warmup Sequence, 80 km/h, 1855 N, 1 Slip Angle sweep in positive and negative directions.

C1: Machine limitations reached on high load step. Tire was unloaded to a safe position during the steering sweep if limitation was reached.

C2: Test aborted prior to last load completion. Will complete last load step in next test.

C3: Test manually aborted by operator due to 120-3Mite surface issues.

C4: Re-test of run 15

Appendix E

Figure E1: Tire Identification Log

Tire ID	Construction Number	Manufacturer	Size	Load Index	Speed Rating	Brand Name	DOT Code	Serial Number / Bar Code
SRTT-AN2209-127	SRTT	Uniroyal	P195/75R14	92	S	Tiger Paw Plus	ANKABBU3004	
5586-01	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25045144
5586-01	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25043873
5586-02	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25044129
5586-02	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR1905	C25040449
5586-03	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25045069
5586-03	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25044128
5586-04	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR1805	C25043877
5586-04	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR1805	C25045066
5586-05	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25045182
5586-06	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25045142
5586-07	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25045067
5586-08	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25043884
5586-09	5D1456Q	Goodyear	P265/70R17	113	S	Wrangler AT/S	4BT6BYDR2005	C25043875

Appendix F

Table F1: Time History Data File Requirements

Header Section

The following data will each be entered on a line within the data file by writing the label as shown in all capital case characters with words separated by a single space. The label is to be separated from the value by a colon (ASCII Code 058). The value associated with the label is then to be written on the same line. The data line is then to be terminated with a carriage return, line feed sequence (ASCII Codes 013 and 010).

Data	Format	Label	Separator	Comments
Testing Organization Name	Character (No Size Limit)	None		
Run Number	Integer	"RUN NUMBER"	Colon (058)	
Data Acquisition Sampling Rate	Integer	"SAMPLE RATE"	Colon (058)	Rate reported as an integer value in units of Hertz (Hz)
Test Date	DD-MMM-YYYY	"TEST DATE"	Colon (058)	DD – day MMM – month abbreviation (JAN) YYYY – four digit year
Tire Size	Character (Length ≤ 25)	"TIRE SIZE"	Colon (058)	Size designation as found on tire sidewall. Do not include service description. Do not include speed symbol characters if they are included in the tire size.
Construction Number	Character (Length ≤ 25)	"TIRE CONSTRUCTION"	Colon (058)	This is a unique identifier for the recipe of the test tire. This number will uniquely identify this tire recipe from all others, even if the tires appear physically the same.
DOT Build Code	Character (Length ≤ 25)	"TIRE DOT CODE"	Colon (058)	The first two characters and the last four digits of the full DOT molded into the tire. This will be the plant code and the build date of the tire.
Manufacturer	Character (Length ≤ 25)	"TIRE MANUFACTURER"	Colon (058)	Tire manufacturer as shown on the tire sidewall.
Inflation Pressure	Integer	"PRESSURE"	Colon (058)	The test inflation pressure during the test. Report the test inflation pressure as an integer value in units of kilopascals (kPa).
Test Rim Designation	Character (Length ≤ 25)	"WHEEL SIZE"	Colon (058)	Test wheel size given as, Bead Diameter Code X Rim Width Example: 18 x 7.0
Fixed Input Angle	Character (Length ≤ 25)	"INCLINATION ANGLE" Or "SLIP ANGLE"	Colon (058)	These lines can continue until all fixed angle inputs are called out for the entire run. Each fixed angle input are to be given in units of degrees with a number format with of 1 decimal location. The value is to be followed by the word "degrees".
Comments	Character (No Size Limit)	"COMMENT"	Colon (058)	A comment of any length can be placed in this location.

Data Section of the File

The main section of this file will be the data. Following the header section described above, the column headers will be written to the file, followed by the column header units, and then the time history data acquired during the tire test.

Data Header Row

The column headers are to be written immediately following the header information above. The column headers are to be exactly those labels provided below. The labels are to be separated by at least one space character (ASCII Code 032), but more spaces can be used. There are not to be any other characters used for spacing of the columns. The line is to be terminated by a carriage return, line feed sequence (ASCII Codes 013 and 010).

Data Header Units Row

The column header units are to be written immediately following the column header row described immediately above. The column header units are to be exactly those labels provided below. The labels are to be separated by at least one space character (ASCII Code 032), but more spaces can be used. There are not to be any other characters used for spacing of the columns. The line is to be terminated by a carriage return, line feed sequence (ASCII Codes 013 and 010).

Data Rows

The time history data are to be written immediately following the column header unit row described immediately above. The characters within the time history data section of the file should only consist of spaces (ASC Code 032), numbers (ASCII Codes 048 thru 057), plus and negative symbols (ASCII Codes 043 and 045), and the line termination characters (ASCII Codes 013 and 010). The time history data are to be separated by at least one space character (ASCII Code 032), but more spaces can be used. There are not to be any other characters used for spacing of the columns. The line is to be terminated by a carriage return, line feed sequence (ASCII Codes 013 and 010).

Channel Name	Column Header	Data Format	Units	Example Data Value
Elapsed Time	ET	F9.3	Seconds	72561.025
Slip Angle	SA	F7.3	Degrees	± 25.698
Inclination Angle	IA	F7.3	Degrees	± 25.698
Surface Speed	V	F7.2	km/h	± 123.56
Spindle Speed	N	F8.2	rpm	± 1256.25
Inflation Pressure	P2	F7.2	kPa	1124.89
Longitudinal Force	FX	F9.2	Nm	± 12345.67
Lateral Force	FY	F9.2	Nm	± 12345.67
Vertical Force (Applied Load)	FZ	F9.2	N	± 12345.67
Overturning Moment	MX	F9.2	Nm	± 12345.67
Spindle Torque	MY	F9.2	Nm	± 12345.67
Aligning Torque	MZ	F9.2	Nm	± 12345.67
Slip Ratio	SL	F9.3	m/m	± 1234.567
Loaded Radius	RL	F6.2	mm	234.56
Tread Surface Temperature Center	TSTC	F7.2	°C	± 123.45, °C
Tread Surface Temperature Center	TSTI	F7.2	°C	± 123.45, °C
Tread Surface Temperature Center	TSTO	F7.2	°C	± 123.45, °C
Surface Temperature	RST	F7.2	°C	± 123.45, °C
Ambient Temperature	AMBTEMP	F7.2	°C	± 123.45, °C

Figure F1: Sample Time History Data File

TEST ORGANIZATION NAME	
RUN NUMBER	: 103
SAMPLE RATE	: 250 Hz
TEST DATE	: 20-Jun-08
TIRE SIZE	: P195/75R14
TIRE CONSTRUCTION	: SRTT
TIRE DOT CODE	: AN2306
TIRE MANUFACTURER	: Uniroyal
PRESSURE	: 241 kPa
RIM WIDTH	: 6.00
INCLINATION ANGLE	: 0.0 Deg
COMMENTS	:

ET (sec)	SA (deg)	IA (deg)	V (kph)	N (rpm)	P2 (kPa)	FX (N)	FY (N)	FZ (N)	MX (N-m)	MY (N-m)	MZ (N-m)	SL (SAE)	RL (mm)	TSTC (degC)	TSTI (degC)	TSTO (degC)	RST (degC)	AMBTMP (degC)
204.278	0.058	0.034	55.890	481.37	243.21	2790.41	248.53	-4521.2	17.67	-54.19	42.59	0.018	299.14	29.5	33.3	28.2	11.1	26.4
204.282	0.059	0.030	55.860	481.09	243.21	2752.51	246.10	-4511.6	15.96	-54.98	39.83	0.018	299.12	29.5	33.3	28.2	11.1	26.4
204.286	0.062	0.036	55.850	480.94	243.07	2709.49	245.64	-4505.5	15.62	-56.68	35.64	0.018	299.11	29.5	33.3	28.2	11.1	26.4
204.290	0.067	0.036	55.830	480.78	243.21	2667.89	242.44	-4505.5	16.98	-57.06	32.31	0.018	299.09	29.5	33.3	28.2	11.1	26.4
204.294	0.062	0.043	55.820	480.58	243.07	2626.71	236.36	-4505.4	17.68	-61.92	28.68	0.018	299.09	29.5	33.2	28.2	11.1	26.4
204.298	0.058	0.033	55.800	480.31	243.21	2581.31	227.74	-4508.6	18.67	-61.30	24.75	0.017	299.12	29.5	33.2	28.2	11.1	26.4
204.302	0.058	0.036	55.790	480.15	243.14	2531.72	223.66	-4512.0	20.95	-61.96	20.50	0.017	299.11	29.5	33.2	28.2	11.1	26.4
204.306	0.061	0.037	55.780	480.00	243.14	2483.82	217.16	-4518.8	23.32	-65.17	17.81	0.017	299.10	29.4	33.3	28.2	11.1	26.4
204.310	0.058	0.034	55.770	479.76	243.07	2435.00	210.41	-4525.8	27.02	-69.58	16.26	0.017	299.10	29.4	33.2	28.2	11.1	26.4
204.314	0.059	0.036	55.760	479.49	243.14	2383.05	202.79	-4531.1	28.97	-71.41	14.49	0.017	299.09	29.5	33.2	28.2	11.1	26.4
204.318	0.059	0.031	55.750	479.26	243.21	2331.90	194.82	-4542.8	33.81	-78.09	14.85	0.016	299.13	29.5	33.2	28.2	11.1	26.4
204.322	0.062	0.037	55.740	478.98	243.21	2276.65	186.52	-4553.0	37.57	-84.43	15.51	0.016	299.08	29.5	33.2	28.2	11.1	26.4
204.326	0.062	0.037	55.730	478.63	243.21	2222.86	175.45	-4560.7	39.95	-93.44	17.05	0.015	299.12	29.5	33.2	28.2	11.1	26.4
204.330	0.061	0.036	55.730	478.40	243.21	2161.59	168.81	-4566.5	41.03	-90.62	19.09	0.015	299.11	29.5	33.2	28.2	11.1	26.4
204.334	0.061	0.036	55.730	478.08	243.14	2103.08	161.19	-4567.6	41.24	-97.60	22.79	0.014	299.17	29.5	33.1	28.2	11.1	26.4
204.338	0.061	0.034	55.720	477.73	243.07	2039.67	153.47	-4575.9	43.75	-103.37	25.77	0.014	299.14	29.4	33.1	28.2	11.2	26.4
204.342	0.061	0.036	55.710	477.42	243.28	1969.48	147.12	-4580.1	44.63	-103.08	27.99	0.013	299.09	29.5	33.1	28.2	11.2	26.4
204.346	0.057	0.034	55.690	477.07	243.07	1901.98	143.85	-4580.0	43.01	-108.86	31.28	0.013	299.12	29.5	33.1	28.2	11.1	26.4
204.350	0.062	0.037	55.700	476.71	243.21	1830.14	141.01	-4581.2	42.85	-111.48	34.19	0.012	299.08	29.5	33.1	28.2	11.2	26.4
204.354	0.062	0.039	55.680	476.36	243.14	1758.11	143.68	-4578.4	41.27	-111.12	36.42	0.011	299.09	29.4	33.1	28.2	11.1	26.4
204.358	0.062	0.040	55.680	476.05	243.21	1683.62	145.25	-4576.9	40.31	-112.68	38.29	0.011	299.07	29.4	33.0	28.2	11.1	26.4
204.362	0.062	0.034	55.670	475.78	243.21	1609.78	148.56	-4573.4	39.49	-113.05	39.99	0.010	299.10	29.4	33.0	28.2	11.2	26.4
204.366	0.062	0.036	55.660	475.43	243.21	1538.19	154.65	-4566.9	37.38	-113.19	41.95	0.010	299.13	29.4	33.0	28.2	11.1	26.4
204.370	0.062	0.036	55.650	475.11	243.21	1465.80	161.82	-4556.0	34.86	-111.39	42.87	0.009	299.11	29.4	33.0	28.2	11.1	26.4
204.374	0.061	0.031	55.650	474.88	243.07	1394.66	165.22	-4547.7	32.70	-115.97	43.98	0.009	299.13	29.4	33.0	28.2	11.1	26.4
204.378	0.058	0.040	55.640	474.53	243.14	1323.10	174.17	-4540.2	30.85	-121.30	43.70	0.008	299.11	29.4	33.0	28.2	11.2	26.4
204.382	0.061	0.036	55.630	474.37	243.21	1251.50	178.74	-4529.9	29.22	-119.81	43.44	0.008	299.07	29.4	33.0	28.2	11.1	26.4
204.386	0.062	0.037	55.640	474.02	243.21	1183.88	182.61	-4514.8	26.04	-126.83	42.79	0.007	299.08	29.4	32.9	28.2	11.2	26.4
204.390	0.059	0.034	55.630	473.86	243.21	1109.91	183.84	-4505.5	23.92	-125.28	40.54	0.007	299.10	29.4	32.9	28.2	11.2	26.4
204.394	0.064	0.037	55.630	473.63	243.28	1037.77	187.89	-4490.7	22.05	-124.04	38.85	0.006	299.08	29.4	32.9	28.2	11.2	26.4
204.398	0.058	0.034	55.630	473.39	243.14	961.64	186.90	-4481.8	20.82	-126.56	35.30	0.006	299.08	29.4	32.9	28.2	11.2	26.4
204.402	0.064	0.033	55.640	473.24	243.28	889.02	184.84	-4478.9	21.47	-127.24	32.23	0.005	299.10	29.4	32.9	28.2	11.2	26.4
204.406	0.055	0.037	55.630	473.08	243.21	811.47	186.69	-4466.2	19.46	-125.07	28.40	0.005	299.10	29.4	32.8	28.2	11.2	26.4
204.410	0.062	0.034	55.640	472.77	243.21	731.54	184.71	-4461.7	21.82	-122.62	24.25	0.004	299.10	29.4	32.8	28.2	11.2	26.4
204.414	0.064	0.033	55.640	472.57	243.21	653.45	180.47	-4459.1	23.37	-122.32	20.66	0.004	299.08	29.4	32.8	28.2	11.2	26.4
204.418	0.061	0.034	55.640	472.42	243.21	574.66	174.16	-4457.7	24.66	-116.63	16.64	0.004	299.08	29.3	32.8	28.2	11.2	26.4
204.422	0.062	0.034	55.650	472.22	243.07	497.88	167.71	-4462.2	27.44	-113.45	13.28	0.003	299.08	29.3	32.8	28.2	11.2	26.4
204.426	0.062	0.041	55.650	472.03	243.01	419.84	161.78	-4465.1	28.72	-105.91	9.70	0.003	299.06	29.3	32.8	28.2	11.2	26.4
204.430	0.059	0.033	55.650	471.87	243.07	345.79	151.25	-4471.2	31.71	-101.57	7.43	0.002	299.05	29.3	32.8	28.2	11.2	26.4
204.434	0.057	0.031	55.650	471.71	243.14	280.54	142.34	-4481.9	34.94	-101.82	6.54	0.002	299.07	29.3	32.7	28.2	11.2	26.4
204.438	0.055	0.033	55.640	471.52	243.14	216.46	130.14	-4491.0	36.02	-104.66	5.80	0.002	299.08	29.3	32.7	28.2	11.2	26.4
204.442	0.070	0.039	55.650	471.36	243.07	157.57	118.41	-4503.4	40.10	-111.32	6.51	0.001	299.03	29.3	32.7	28.2	11.2	26.4

Appendix G: New Test Surface Preparation

G1 Remove old surface by separating the adhesive backed test surface from the flexible steel belt. This should be done with the steel belt installed on the test machine to avoid damaging the belt. If this is not possible, then support the area of the belt where the surface is being removed.

G1.1 A sharp putty knife should be used to aid removing the surface while pulling the adhesive backed test surface from the belt.

G1.2 A heat gun may be used to weaken the adhesive backing on the 3M-ite material, aiding the removal from the steel belt.

G1.3 Solvents may also be used to aid in the removal of the remaining adhesives from the steel belt. The heat gun should not be used in combination with the solvents as they are typically highly volatile and flammable.

G2 Install new surface once the entire old surface, including all adhesives, has been removed.

G2.1 Scribe a straight line around the belt between 2½ and 3 in from the edge of the belt. This line should be as straight as possible. A felt tip marker may be used for this task.

G2.2 Apply the new surface to the steel belt by laying the rolled surface on the belt, using the scribed reference line as a guide.

G2.3 Begin removing the protective backing from the surface, carefully following the scribed reference line.

G2.4 Rotate the steel belt on the machine as necessary to facilitate the application of the surface over the entire length of the belt by manually rotating the drums at either end of the machine.

G2.5 Once the length of the steel belt has been covered with the 3M-ite material carefully cut the surface to match the beginning cut above, avoiding gaps and overlapping material.

G2.6 The new surface must now be pressed onto the steel belt to properly secure it for testing. The adhesive is pressure sensitive and, therefore, must be pressed onto the steel belt surface.

G2.6.1 The specific tire used for the pressing the surface onto the steel belt in is not important. A tire such as a high performance, low profile, wide tread width tire used for can be used for this purpose.

G2.6.2 Allow the machine to operate under those conditions for approximately 30 minutes until most of the air bubbles between the surface and the steel belt are removed.

G3 Break-in the new surface using the mason's stone to dull the grit of the newly applied 3M-ite test surface.

G3.1 Scrub the entire roadway surface using a mason's stone.

G3.2 Apply the mason's stone to the surface with the length of the stone perpendicular to the direction of travel for the surface.

G3.2.1 Use approximately 90 to 130 N (20 to 30 lbs) of force downward on the stone, evenly across the stone.

G3.2.2 Cover the new test surface with 10 passes under the stone.

G3.2.3 Continue this process by moving the stone over to the next area of the test surface that has not been dulled by the stone.

G3.2.4 Continue the above process until the entire surface has been dulled by the stone.

G4 The surface must now be cleaned and validated using the surface cleaning procedure given in Appendix I.

G5 All control tires are to be tested to verify the condition of the new test surface.

G6 Record in the Equipment Maintenance Log and in the Force and Moment Test Log regarding the test surface replacement.

Appendix H: Test Surface Cleaning Procedure

H1 Some tires will deposit an oily residue on the test surface while being tested. This cannot be avoided during the test for those tires as it is an artifact of their tread compounds. However, before testing tires for the next Tire Test Request, the test surface must be cleaned to ensure consistent and repeatable test results.

H2 The specific tire used for the cleaning is not important. The tire selected should be a wide tire and itself one that does not deposit an oily residue on the test surface. A high performance, low profile, wide tread width tire is typically used for this purpose.

H3 The machine will be manually run until the operator visually determines the oily residue has been removed from the test surface.

H3.1 Set the following machine control parameters to the indicated values.

Speed	7.6 kph
Load	3500 N
Camber angle	0 degrees

H3.2 The slip angle is controlled using a function generator to drive the machine's slip angle set point.

Average set point	0 degrees
Frequency	0.1 Hz
Amplitude	8 degrees
Shape	Sine Wave

H4 The test surface cleaning should be followed retesting the previous ASTM E1136 SRTT, verifying the condition of the test surface per paragraph 4.3.1.

H5 Record in the Equipment Maintenance Log and in the Force and Moment Test Log the test surface was cleaned.

Appendix J

Machine Capacities

J1 Minimum Machine Capability. The machine described by the following list of requirements is the minimum capability required to conduct this testing. Equipment meeting these requirements will be able to perform tests to this procedure for most passenger car vehicles. The limits of this equipment will be reached often during the execution of this procedure on large passenger car tires and light truck tires appropriate for the larger vehicles within General Motors vehicle line.

J2 Full Machine Capability. The machine described by the following list of requirements will be capable of executing this procedure on the majority of tires used with the General Motors vehicle line.

Table J1: Minimum Equipment Capability

Parameter	SAE Symbol	Full Scale	Units
Longitudinal Force	F_X	± 15	kN
Lateral Force	F_Y	± 15	kN
Radial Load	F_Z	-25	kN
Overturning Moment	M_X	± 10	kN-m
Spindle Torque	M_Y	≥ +2	kN-m
Aligning Moment	M_Z	± 3	kN-m
Loaded Radius	R_L	Min. ≤ 250 Max. ≥ 450	mm
Slip Angle	α	± 25	Degrees
Inclination Angle	γ	± 10	Degrees
Speed	V_S	80	km/h
Maximum Speed (Reduced Capability)	V_S	>80	km/h
Spindle Speed	N_S	± 1200	rpm
Slip Angle Rate		>10	Degrees/s
Inclination Angle Rate		>4	Degrees/s
First Natural Freq (structural and load cell)		>70	Hz

Table J2: Full Equipment Capability

Parameter	SAE Symbol	Full Scale	Units
Longitudinal Force	F_X	± 25	kN
Lateral Force	F_Y	± 30	kN
Radial Load	F_Z	-30	kN
Overturning Moment	M_X	± 15	kN-m
Spindle Torque	M_Y	Min. ≤ -10 Max. ≥ +7	kN-m
Aligning Moment	M_Z	±3	kN-m
Loaded Radius	R_L	Min. ≤ 250 Max. ≥ 550	mm
Slip Angle	α	± 30	Degrees
Inclination Angle	γ	± 10	Degrees
Speed	V_S	80	km/h
Maximum Speed (Reduced Capability)	V_S	>80	km/h
Spindle Speed	N_S	± 1200	rpm
Slip Angle Rate		>90	Degrees/s
Inclination Angle Rate		>38	Degrees/s
First Natural Freq		>100	Hz